

ABDALLAH SHEE

Full-Stack Software Engineer

Node.js · Express · Next.js · TypeScript · PostgreSQL · Python (FastAPI)

Nairobi, Kenya | +254 796 515 302 | abdallahshee664@gmail.com

Portfolio: <https://portfolio-git-main-abdallahshees-projects.vercel.app/>

PROFESSIONAL SUMMARY

I'm a Full-Stack Software Engineer who builds end-to-end with Node.js, Express.js, and Next.js, using TypeScript across the client, server, and database layers, with PostgreSQL powering scalable, type-safe data storage. I also build high-performance APIs in Python with FastAPI, giving me flexibility across both ecosystems. Beyond the code, I focus on understanding the business problem first, then modeling systems that actually meet it — an approach that's shaped every production application I've shipped, backed by solid API design, authentication and authorization, clean middleware architecture, and backend systems tuned to perform at scale.

CORE STRENGTHS

High-Performance API Design with FastAPI

Builds self-documenting, production-grade APIs in Python with FastAPI — Pydantic validation, custom middleware, and async endpoints deliver speed, security, and Swagger docs that stay in sync with the code automatically. This means less time writing docs by hand and fewer integration bugs, since anyone consuming the API always sees an accurate, up-to-date contract.

Type-Safe Frontend Architecture

Ships fast, reliable UIs with React, TypeScript, TanStack Router, and TanStack Query — with Zod catching bad input before it becomes a bug, and Tailwind CSS with DaisyUI turning designs into polished interfaces quickly. The result is a frontend that's both fast to build and resilient to the runtime errors that slip past looser type systems.

Business-Driven System Design

Translates business goals into working software by modeling data and logic around real workflows, not assumptions — the foundation behind a regulatory-compliant invoicing platform and multi-tenant SaaS systems built to actually meet their objectives. This focus on the "why" behind a feature keeps solutions aligned with what stakeholders actually need, not just what's technically convenient to build.

TECHNICAL SKILLS

- **React** — component-driven UI architecture with hooks, context, and lifecycle management, paired with Redux, TanStack Query, TanStack Router, and TanStack Start for predictable state management and smooth client-side navigation
- **TypeScript** — advanced, end-to-end across client, server, and database layers; type-safe contracts, generics, and shared types throughout the stack
- **Next.js** — App Router, SSR/SSG/ISR, Server Actions, Edge Middleware, and route-level data fetching for fast, SEO-friendly applications
- **Node.js & Express.js** — RESTful and real-time API design, middleware pipelines, modular service architecture, and structured error handling
- **Python & FastAPI** — REST API development, including routers, dependency injection, request/response validation with Pydantic, and async endpoints for high-throughput services
- **PostgreSQL** — schema design, type-safe queries, migrations, performance tuning, and optimized search/filtering for high-traffic platforms
- **Additional Data Stores:** Redis and AWS ElastiCache for caching, session storage, and reducing database load under heavy traffic
- **Authentication & Authorization:** session/token-based auth, OAuth flows, role-based access control (RBAC), secure middleware and request/response validation
- **Cloud & Serverless:** Vercel, AWS Lambda, edge middleware, horizontally scalable stateless service design, CDN/edge-caching strategies
- **Tooling & Collaboration:** Git/GitHub (branching, PR workflows, GitHub Actions/CI-CD), npm/pnpm, Agile cross-functional collaboration
- **AI-Assisted Development:** agentic coding tools including Claude Code for code generation, refactoring, debugging, and test scaffolding

PROFESSIONAL EXPERIENCE

Full-Stack Developer — Quantum Plus Ltd

Jun 2023 – Sep 2025

Mombasa, Kenya

- Built and maintained full-stack, production-grade applications using Next.js, Node.js, Express, TypeScript, and PostgreSQL, supporting multi-user SaaS and e-commerce platforms.
- Designed and developed secure, type-safe REST APIs with Express.js middleware layers and scalable backend architecture to support business growth.
- Implemented role-based access control (RBAC) to power secure, multi-tenant SaaS and e-commerce functionality across distinct user roles and permission levels.
- Drove testing, debugging, and PostgreSQL query/performance optimization to ensure system reliability, data integrity, and compliance across customer-facing features.
- Streamlined frontend state management and data-fetching workflows using Redux, Zustand, and TanStack Query, improving UI responsiveness and reducing boilerplate across the codebase.
- Collaborated with product managers, designers, and QA in an Agile workflow, using Git/GitHub PR reviews and CI/CD pipelines to ship features reliably and on schedule.

Data Analyst — CloudFactory Kenya

Feb 2022 – Mar 2023

Nairobi, Kenya

- Used Python to process, clean, and structure large-scale datasets, building repeatable automation scripts that powered reliable downstream software and analytical workflows.
- Designed and built Python-based data pipelines and preprocessing scripts to support programmatic data handling and machine learning workflows.
- Built internal FastAPI services to expose processed datasets and automation results to downstream tools, applying Pydantic models for request/response validation.
- Developed and tested predictive models and algorithms in Python following scalable, maintainable coding practices, strengthening foundations in data-driven, production-quality engineering.

Junior Web Developer — Swahilipot Hub Foundation

2020 – 2022

Mombasa, Kenya

- Built responsive, interactive UI components using React and TypeScript, translating UI/UX designs into functional, cross-browser-compatible code.
- Collaborated closely with designers and backend developers to integrate APIs and deliver cohesive end-to-end features.
- Managed database integration with PostgreSQL to support reliable CRUD operations across core application features, laying the groundwork for full-stack development expertise.

EDUCATION

Bachelor of Business Information Technology

Dedan Kimathi University of Science and Technology, Nyeri | 2014 – 2018

Hands-on experience in programming, web development, database management, and the software development lifecycle, with foundational knowledge in business information systems and IT infrastructure.

CERTIFICATIONS

- Nanodegree in Cloud Computing (AWS) — Udacity, 2023
- Certificate in Financial Technology (FinTech) — Corporate Finance Institute (CFI), May 2023
- Certificate in Business Intelligence — Corporate Finance Institute (CFI), Nov 2022
- Certificate in Software Development — Power Learn Project Academy, Nairobi, Jan 2022 – Apr 2022